

HVIA Data Analysis Training Task

Olist Business Discovery | Final Integrated Report (Tasks 1-6) | A + B + C + E

Reading path: A Company -> B Dataset -> C Analysis (evolved with Tasks 2-6) -> E HVIA Solutions (+ E.8 extensions). Descriptive evidence only - no causal claims.

A. Company Research

Olist helps Brazilian SMB retailers sell more through technology. Below: who they are, how money worked in the dataset era, what was added later, then a dated timeline. The public order extract (Kaggle ~2018) reflects the marketplace aggregator / Olist Store model - not today's full SaaS + fintech stack. Keep that split while reading B and C.

Executive Summary

Olist's 2016-2018 marketplace-aggregator story, as the data shows it: fast order growth on a shared storefront where a small share of late deliveries coincides with much lower review scores, customers rarely return, and revenue is concentrated in a thin seller elite. The public order extract (roughly 2016-09-04 to 2018-10-17) describes that aggregator model - not today's SaaS + fintech stack.

Strongest descriptive signals: only 6.8% of delivered orders arrive after the ETA, yet average review scores show 4.29 (on-time/early) vs 2.27 when late; crude one-time buying is 96.88% (fixed-window H1-2017 12m repeat is 4.68% on eligible customers); the top 10% of sellers account for ~67.5% of product revenue; and late product GMV is R\$985,924.34 (7.25% of GMV), concentrated in the top revenue decile - with any commission math shown as ASSUMED scenarios only.

Tasks 2-6 add: delivery-stage timing (handoff and transit gaps between late and on-time orders), seller exposure tiers, a category-risk lens restricted to categories with >=100 delivered orders, fixed-window cohort retention with censoring labeled, and late-GMV exposure sizing by seller and category. Integrated direction: predict late risk before the ETA is missed, rank sellers by health rather than GMV alone, protect the first-order experience, and size late-GMV exposure as labeled scenarios for investigation.

A.1 Snapshot

Field	Detail
Legal name	Olist Serviços Digitais Ltda. (CNPJ 18.552.346/0001-68)
Founded	2014 (commercially active from 2015; reported; sources differ on 2014 vs 2015)
Founder & CEO	Tiago Dalvi
HQ	Curitiba, Paraná, Brazil (+ São Paulo, Bento Gonçalves)
Team size	~790-850 (approximate; after SaaS-oriented restructuring)
Valuation / funding	\$1.5B unicorn (Dec 2021, reported) ~\$300M+ funding (approximate; unverified)
Merchants / scale	50,000+ merchants (reported) R\$60B+ ecosystem invoicing (company-reported)
Primary customer	SMB retailers / brands (sellers); end consumers are demand-side in the CSVs

A.2 Dataset-era model (2016-2018) - what the data describes

Flow: small seller joins Olist -> products listed under Olist's verified storefront on major marketplaces -> customer buys from "Olist" -> seller fulfills. Monetization: roughly 19-21% blended commission (reported/unverified; C.6 take-rates are ASSUMED scenarios) plus modest seller subscription / membership fees.

- Thin intermediary margins: historically gross margins in a mid-30s percent range were hard to expand while operating the aggregator layer.
- Reputation contagion: a few late or poor sellers could damage the shared Olist account rating on Mercado Livre, Amazon, etc., hurting thousands of good sellers at once.
- Merchant churn at scale: as seller GMV grew, a ~20% take-rate pushed sellers to open direct marketplace accounts; platforms also made direct onboarding easier over time.

A.3 Later model (context only) - what was added after the dataset

Olist evolved into a retail technology ecosystem often framed as a "Commerce OS": Tiny ERP, Vnda (D2C), Olist Pax (logistics), Flip (credit), PDV, and Lis (embedded AI). Indicative later mix (unverified estimate): ~65% recurring SaaS, ~20% fintech/credit, ~10% logistics spread. Use this for company research; do not force 2025-2026 products into every 2016-2018 KPI.

- Competitors (examples): Bling, Nuvemshop, VTEX, Melhor Envio, Magazord - different ERP / storefront / shipping mixes.
- Stack (high level): Python/Go/React; Kafka; PostgreSQL/Redis/Elasticsearch; AWS + Spark/Databricks-style analytics.

A.4 Sequential timeline - what was added when

Read top to bottom. The 2016-18 band is locked to the public CSV extract. Later rows explain strategic evolution; they are not additional columns in the order tables.

2007	Solidarium (reported): physical retail network supporting artisans and small producers (founder Tiago Dalvi's pre-Olist chapter - same SMB DNA).
2014-15	Olist founded. Insight: SMBs struggle to reach customers on major marketplaces, not only to open a store. Olist launches as a tech + logistics bridge / aggregator.
2016-18	DATASET WINDOW (this task). Olist Store / shared storefront: sellers fulfill under one Olist face; commission ~19-21% (reported/unverified) + fees; ~99k public orders. Shared reputation is the structural risk. All Sections B-C interpret operations in this window.
2019-20	Ecosystem build: Clickspace (marketplace / social commerce infrastructure) and PAX Logtech (later Olist Pax - carrier aggregation, rate shopping, discounted labels).
2021	Adds Tiny ERP (catalog, inventory, Brazilian tax invoicing) and Vnda (D2C / headless storefront). Unicorn at \$1.5B (Dec 2021). Software becomes a clearer growth path.
2022-24	Tech-winter discipline: cost control, focus on higher-margin SaaS recurring revenue, and operating leverage versus pure marketplace take-rate economics.
Aug 2025	Adds fintech Flip -> Flip by Olist for merchant working capital, supported by an FIDC credit structure (~R\$90M, reported). Underwriting edge (hypothesis): ERP sales/invoice signals.
2026	Aug: reported operational breakeven path. Sep: CEO announces wind-down of the shared marketplace storefront model and a full shift toward SaaS + fintech - closing the chapter that the 2016-2018 dataset documents.

A.5 Takeaway - how to use this in B and C

Acknowledge today's Commerce OS in company research. For dataset understanding and analysis, stay inside 2016-2018: shared reputation, commission on orders, seller-fulfilled shipping, freight and ETA as customer-visible promises. Early dataset signals already point to late delivery associated with lower reviews, near-zero repeat purchase, and seller revenue concentration - expanded next.

Bridge KPI (from CSVs)	Signal
~99.4k orders / ~96k unique	Scale of the public extract
Late vs ETA 6.8% with reviews 4.29 vs 2.27 (descriptive)	Reputation risk > volume risk
Crude one-time 96.88% (H1-2017 12m repeat 4.68%)	Acquisition-led growth, weak LTV
Top 10% sellers -> ~67.5% product revenue	GMV + quality concentration

B. Dataset Understanding

Public Brazilian E-Commerce extract (Olist Store). Purchase timestamps span roughly 2016-09-04 to 2018-10-17. Nine related CSVs. Not in scope: marketplace ad spend, seller subscription invoices, or post-2018 SaaS / Flip product telemetry.

B.1 Nine tables (structure)

Table	Rows / role
orders	99,441 Hub: status + purchase/approval/carrier/delivery/ETA
order_items	112,650 Line items: product, seller, price, freight (~1.14/order)
customers	99,441 Location; customer_unique_id = true person (96,096)
sellers	3,095 Merchant location
products	32,951 Category, photos, weight/dims
payments	103,886 Type, installments, value (some split payments)
reviews	99,224 Score 1-5 + optional text (~1 per order)
geolocation	1,000,163 Lat/lng; many rows per ZIP -> mean before distance
category_translation	71 Portuguese category -> English

B.2 How to join (sequential)

Step 1 | Start at orders

Filter/status first. ~97% delivered. Delivery KPIs need delivered + non-null delivered_customer_date (96,470 orders in the SLA cut).

Step 2 | Attach items -> products -> sellers

Revenue and freight live on items; aggregate to order before AOV or late-rate. Category via products + translation: 610 null category + 2 unmapped PT names -> Unknown bucket (1,559 delivered items / R\$185k revenue); quantified for totals reconciliation, excluded from ranked headlines.

Step 3 | Attach customers + payments

Retention only via customer_unique_id. Payments: ~74% credit card, ~19% boleto, ~6% voucher; ~3k orders have multiple payment rows.

Step 4 | Attach reviews + geo

Aggregate review_score if duplicate order_id (551 duplicate rows). ZIP geo is not unique - mean lat/lng per prefix before seller-customer distance.

B.3 Data quality + exploration snapshot

Main traps are grain mismatches (item vs order) and geo duplication - not corrupt CSVs. Review titles missing ~88% and messages ~59%: scores scale; text themes are biased toward customers who wrote comments.

KPI	Value
Product revenue / freight / AOV	R\$13.6M R\$2.3M ~R\$138
Late vs ETA (delivered)	6.8% of 96,470
Review on-time vs late	4.29 vs 2.27 stars (descriptive)
One-time buyers	96.88% crude (H1-2017 12m repeat 4.68%)
Top 10% sellers -> revenue	~67.5%
SP share (customers / sellers)	~42% / ~60%
Cancel + unavailable	~1.24% (secondary vs late & loyalty)

Takeaway: order-centric marketplace ops data. Fulfillment reliability and seller heterogeneity sit on a one-shot customer base - the chain of associations in Section C.

C. Analysis & Business Story | Chain of associations

Read as a sequence of associations: growth stress -> logistics and geography -> late deliveries -> review collapse and written complaints -> weak repeat purchase - amplified by a concentrated seller base under one shared Olist face. Tasks 2-6 add stage timing, seller/category concentration, cohort windows, and late-GMV exposure sizing. Note: Task D is folded into C (no separate D section).

C.1 Growth co-occurred with operational pressure

Orders scaled fast (Jan 2017 -> Jan 2018 YoY ~+809%, with an early-base caveat: Jan 2017 had only ~800 orders). Peak month Nov 2017: 7,544 orders (Black Friday context). Cancel + unavailable together ~1.2% - secondary versus lateness and loyalty. Peaks coincide with wider SLA gaps (descriptive; not proof peaks cause lateness).

New evidence (late-GMV seasonality, see E.8): within-month late product GMV share spikes at Nov-2017 12.49%, Feb-2018 15.05%, and Mar-2018 19.03% (item grain; locked is_late broadcast) - peaks co-occur with volume stress and warrant investigation as seasonal exposure, not proof of a single root cause.

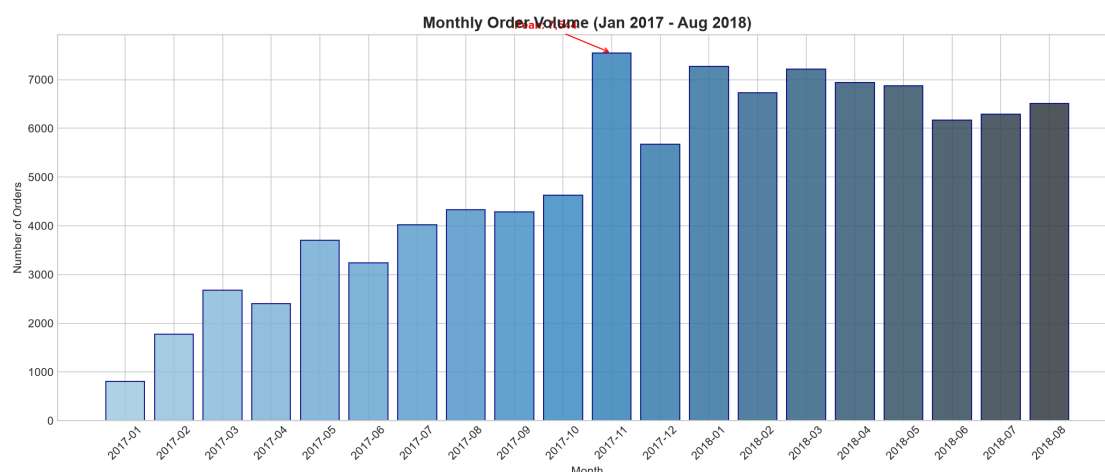


Fig C.1 (Chart 01) Monthly order volume - growth and seasonal spike.

C.2 Distance and freight coincide with an uneven promise

LATE GRAINS (same locked is_late = delivery delta_days > 0): order 6,534 / 96,470 | item-broadcast 7,264 (x1.112) | seller-pair 6,547 (97,811 pairs; 1,278 multi-seller orders with items, of which 1,275 delivered). Different grains, not an error.

São Paulo holds ~42% of customers and ~60% of sellers. Same-state delivery averages ~7.5 days versus cross-state ~14.7 (+~7.2 days). Median seller-to-customer distance ~434 km (distance <-> delivery days $r \sim 0.39$). Freight totals ~R\$2.3M (~17% of product revenue); at item level, freight is ~21% of (price + freight).

Glossary (first use): New evidence in this report: delivery stages; seller exposure tiers; category-risk lens; late-GMV view (detail in E.8). D1 = top revenue decile (310 sellers; a ranking label, not a problem ID). Broadcast=order late flag copied to each item. Locked is_late=delivery delta_days > 0 on 96,470 delivered. Grain=unit of counting (order / item / seller-pair / customer).

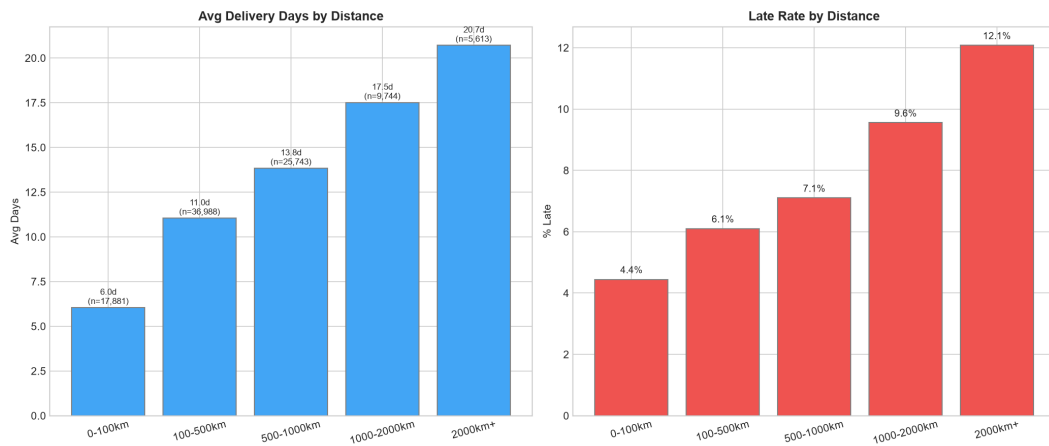


Fig C.2a (Chart 16) Distance vs delivery - farther pairs show longer times and higher late shares (descriptive).

New evidence - Delivery stages (see E.8): Full-chain n=96,455. Medians (≥ 0): approval 0.34h / handoff 44.38h / transit 7.10d / total 10d. Late vs on-time gaps: +0.07h / +30.57h / +19.25d / +22d. p90 handoff ~144.55h. DQ negatives flagged (not dropped): handoff 1,350 (1.40%), transit 23. Shows where time sits; cannot establish which stage causes lateness.

Delivery stage breakdown | Full-chain n=96,455 of delivered 96,470
Negatives: handoff=1,350 (1.40%), transit=23 (0.024%) | Stats on ≥ 0 (unclipped); plot clipping labeled (not winsorized)

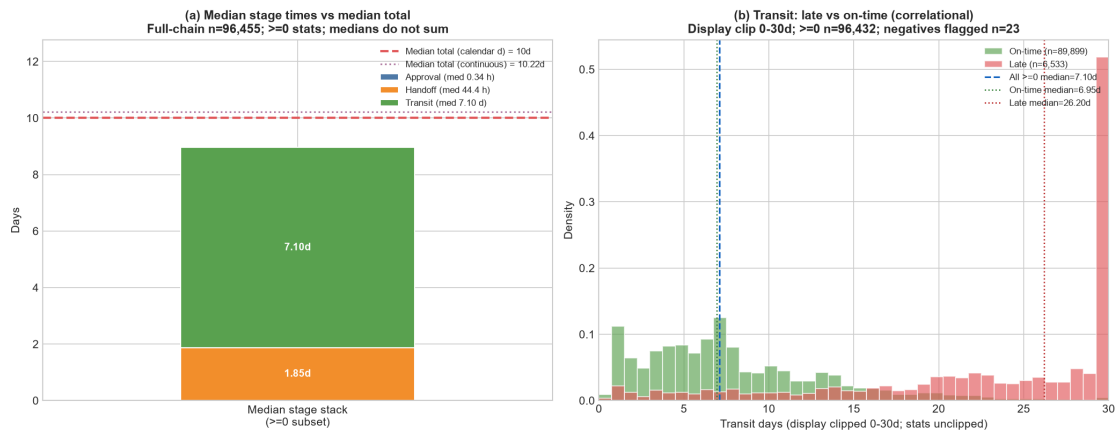


Fig C.2b (Chart 19a) Most delivery time sits in transit; late orders show far longer transit and handoff tails. Technical detail: full-chain n=96,455; medians on ≥ 0 subsets; panel (b) transit density late vs on-time; correlational; see grain definitions in C.2.

New evidence - Category freight lens (see E.8; ≥ 100 delivered orders only; 51/71 cats = 97.77% items / 97.63% revenue): electronics mean item freight ratio 35.81%; audio late rate 11.60% (362 items / 348 orders). dvds_blu_ray 40.37% freight at n=56 is sensitivity-only (below ≥ 100 headline). Freight ratio vs late rate $r=0.12$ (descriptive); vs review $r=0.03$ - co-occurrence is weak, not causal.

Category $\times$ risk | Primary grain = delivered ITEM row (is_late broadcast from locked order rule)
Revenue = all-items SUM(price) reconciles to platform R\$ 13,591,644 | Threshold ≥ 100 delivered orders | Unknown excluded from ranks | Correlational only

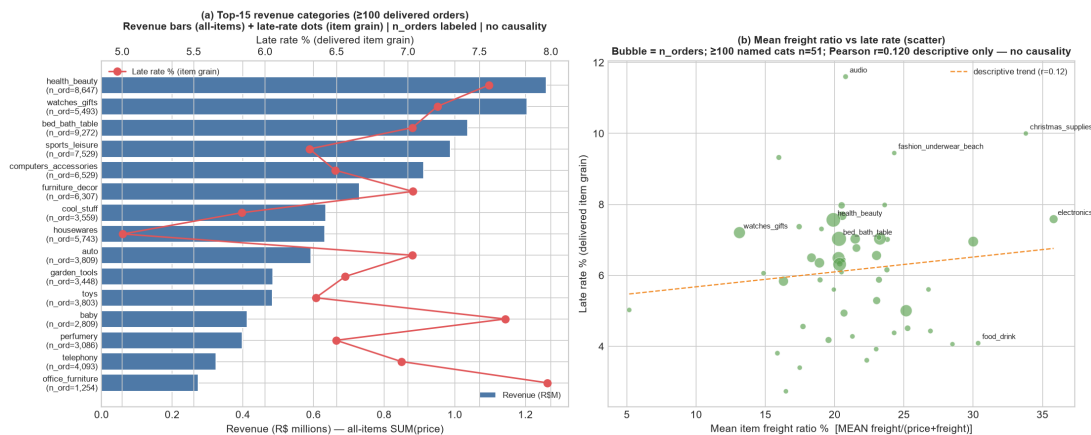


Fig C.2c (Chart 21b) Freight cost does not show a strong relationship with lateness in this descriptive cut. Technical detail: item grain; ≥ 100 delivered orders; n labeled; Unknown excluded; $r=0.12$; descriptive only. Panel (a) top-15 revenue + late dots.

C.3 Late delivery is associated with lower shared-account scores

Note on seller counts: the floor-based top-10% count (309 sellers, 67.49%) differs slightly from the qcut-based D1 decile (310 sellers, 67.56%) used in Task 3-6 seller tables - both round to ~67.5%; D1 is the primary reference from here on.

Only 6.8% of delivered orders arrive after the ETA, but average review score is lower when late: 4.29 (on-time/early) vs 2.27 when late (-2.02 stars, descriptive). Mean actual delivery ~12.1 days versus mean ETA ~23.4 - a large buffer exists, yet late cases still show lower scores. Under aggregator logic this pattern is consistent with reputation contagion on the shared marketplace account, not a CX footnote. Among 1-star comments, delivery/delay keyword hit-rate (unvalidated) ~41% versus product/quality ~13% (commented 1-stars only; n=8,744).

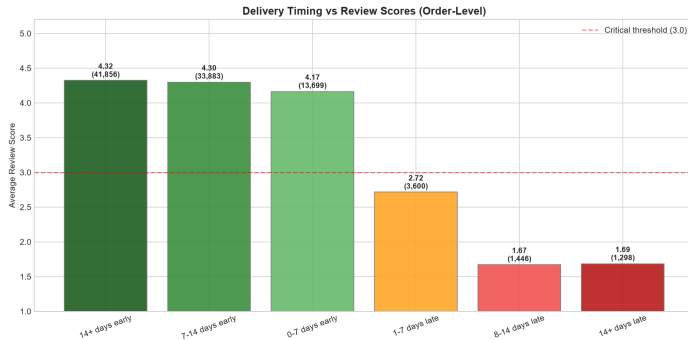


Fig C.3a (Chart 06) Late vs on-time reviews

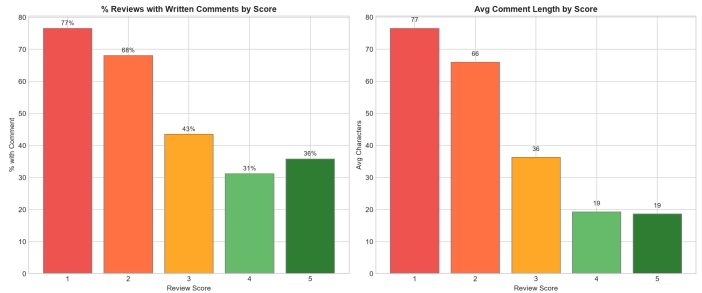


Fig C.3b (Chart 18) 1-star text themes (keyword hit-rate)

New evidence - Seller + stage concentration (see E.8): D1 (top revenue decile, 310 sellers): 67.56% of product revenue (R\$9,183,174.76) and 63.14% of seller-pair late flags. Top-30 sellers by late count hold ~30.1% of pair-late. Operational coverage sellers >=10 delivered: 1,237 = 93.9% volume / 90.2% revenue. Stage gaps (handoff +30.57h, transit +19.25d) co-occur with the locked late flag - warrants investigation, cannot establish causality.

Stage detail: see Fig C.2b (Chart 19) - transit density for late vs on-time orders.

Seller concentration of risk | Revenue deciles on n=3,095 sellers (zero-delivered isolated n=125)
Delivered seller-order pairs=97,811 on locked base 96,470 | Correlational only — no causal seller attribution

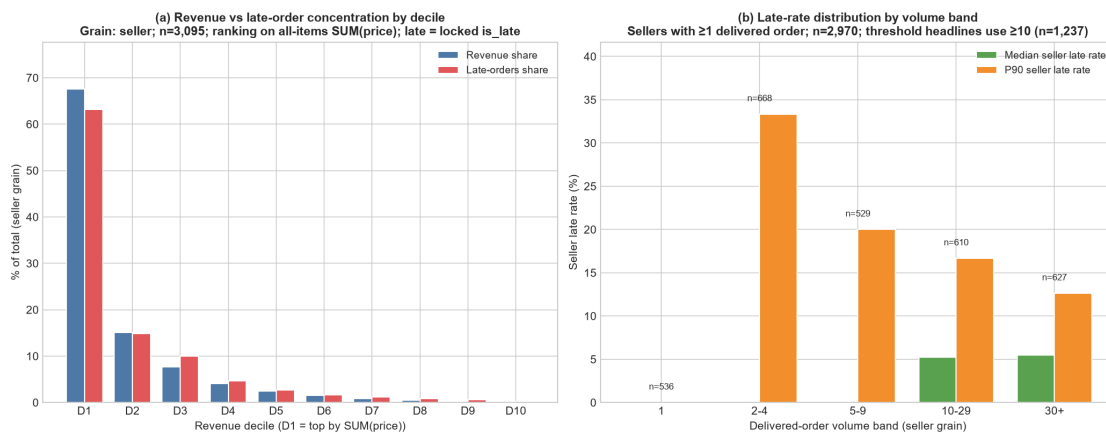


Fig C.3c (Chart 20a) Revenue and late flags are concentrated in the top decile, while late rates stay flat across deciles. Technical detail: n=3,095 sellers; seller-pair grain; locked is_late; descriptive only.

C.4 Almost no loyalty + a thin seller elite

Of 96,096 unique customers, crude one-time 93,099/96,096 = 96.88% (~96.9%) buy once and only ~3.1% return (2+ orders) in the full window. Fixed-window counterpart: H1-2017 12m repeat 666/14,239 = 4.68% (i.e. ~95.3% did not repeat within 12m; eligible 29,087; censored 67,009). Commission LTV would barely compound if repeat stays low (hypothesis; no CAC or customer-value data in scope); a weak first delivery experience coincides with lower likelihood of order #2 (hypothesis, not proven). Among 3,095 sellers, the top 10% drive ~67.5% of product revenue while the bottom 50% contribute ~3.2% - GMV dependence and long-tail reputation risk share the same Olist storefront.

Cohort window (Task 5): Crude full-window one-time 93,099/96,096 = 96.88% remains locked alongside H1-2017 12m repeat 666/14,239 = 4.68% (71 with 3+; median days to second 60.5d). Eligible denominator 29,087; right-censored 67,009 (first_date after 2017-10-17). Different windows/denominators - not a contradiction.

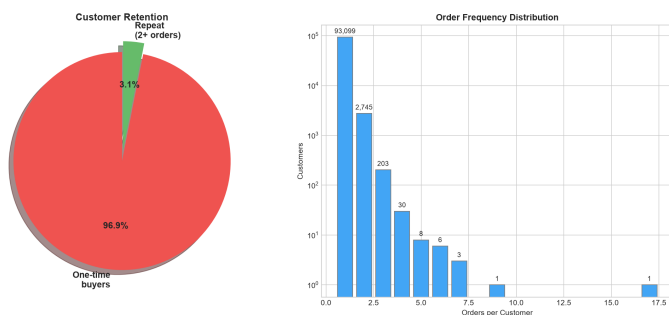


Fig C.4a (Chart 08) One-time vs repeat buyers

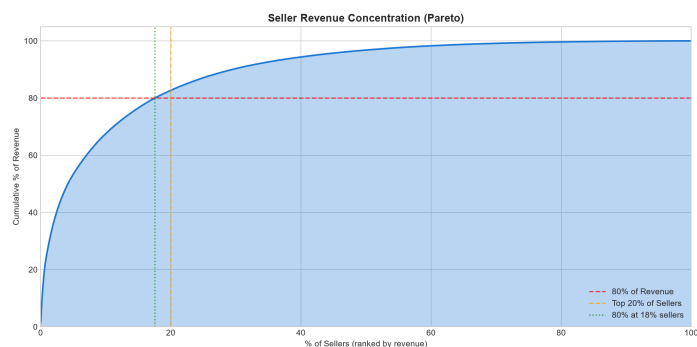


Fig C.4b (Chart 10) Seller revenue Pareto

Cohort retention | Customer grain = customer_unique_id | Fixed 365-day window from first purchase | Eligibility cutoff 2017-10-17 | H1 n=14,239 | Descriptive only — no causality

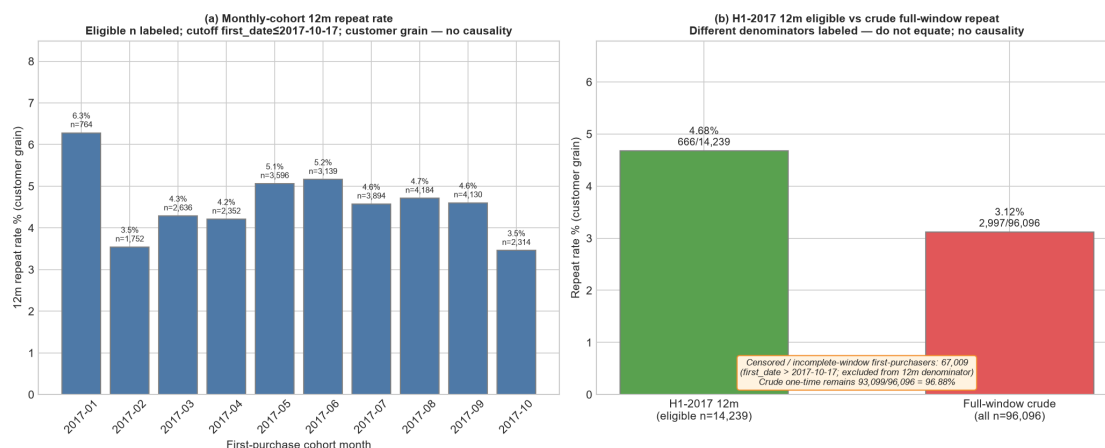


Fig C.4c (Chart 22) Fixed-window repeat stays low even with full observation time. Technical detail: customer grain; eligible n labeled; censored 67,009 annotated; no causality.

Descriptive: first-order lateness vs 12m repeat (eligible cohort, no causality): Eligible 29,087 (first_date ≤ 2017-10-17): on-time first 1,226/26,860 = 4.56%; late first 35/1,010 = 3.47% (-1.10 pp gap, descriptive); non-delivered/unclassifiable first 57/1,217 = 4.68%. Total eligible repeat 1,318/29,087 = 4.53%. Cross-foots H1 666/14,239 = 4.68%. Delivered-first only; same locked is_late; fixed 365d window.

New evidence (exposure concentration, see E.8): D1 holds 67.56% revenue and 63.14% of seller-pair late flags; top-30 late sellers ~30.1% of pair-late - creates exposure at the elite and warrants tiered investigation, not a claim that sellers cause lateness.

C.5 Priority spine (what to solve next)

Priority order using the same P1-P7 labels as the register below (* = new evidence from Tasks 2-6 attached; detail in E.8). P6 feeds P1. Language remains correlational (shows / co-occurs / creates exposure).

P#	Problem -> business stake (2016-18 model)
P1*	Late delivery associated with review drop shared marketplace reputation
P2*	Near-zero repeat purchase acquisition-cost hypothesis, weak LTV (no CAC data)
P3*	Seller Pareto + long tail GMV fragility + contagion
P4	Geo / distance SLA gap uneven national promise
P5*	Freight burden price perception + delay sensitivity
P7*	Growth/seasonality pressure + late-GMV exposure seasonal volume stress on SLA (scenario commission)

Problem register P1-P7 (Scale + New evidence)

P#	Problem	Scale	New evidence / Limitation
P1	Late delivery <-> review collapse	6,534/96,470=6.77%; reviews 4.29/2.27	Stages: handoff +30.57h, transit +19.25d (full-chain 96,455). Limitation: correlational; 8.11% sensitivity only.
P2	Near-zero repeat / loyalty	Crude one-time 96.88% (93,099/96,096)	H1 12m 4.68% (666/14,239); median 60.5d; censored 67,009. Limitation: windows differ; no LTV proof.
P3	Seller Pareto + quality mix	Top 10% ~67.5% rev; n=3,095	D1 67.56% rev + 63.14% pair-late; top-30 ~30.1%; >=10: 1,237 (93.9% vol / 90.2% rev). Limitation: not inefficiency
P4	Geo / distance SLA gap	Same-state ~7.5d vs cross ~14.7d; r~0.39	Unchanged primary; stage/geo still descriptive. Limitation: distance is a proxy, not carrier ID.
P5	Freight burden	~14.2% of paid; ~17% of product; mean-item ~21%	electronics 35.81% (>=100); dvds 40.37% n=56 sensitivity-only; freight-late r=0.12. Limitation: weak
P6	Review-text delivery anger	~41% delivery keyword hit-rate (unvalidated) in commented 1-stars	Unchanged primary text scan. Limitation: commented subset + unvalidated stems.
P7	Growth / seasonality pressure	Nov-17 peak 7,544 orders	Late-GMV monthly spikes Nov-17 12.49%, Feb-18 15.05%, Mar-18 19.03%. Limitation: seasonality context, not causal.

Bridge to HVIA solutions (next section): start from problems 1-3 - predictive delay / ETA, seller SLA risk scoring, and first-order experience protection - not from a tool list. E.8 extends cores with delivery-stage, exposure-tier, category-risk and late-GMV extensions plus a P2 60d refinement.

C.6 Financial Exposure (Task 6)

ASSUMPTION BANNER: Take-rates 15% / 19% / 21% are ASSUMED scenarios only - NOT loss, NOT verified take-rate, NOT recoverable revenue.

Product GMV = SUM(price) = R\$13,591,643.70. Late product GMV (is_late broadcast to items) = R\$985,924.34 = 7.25% of GMV. D1 late GMV R\$680,097.81 = 68.98% of late GMV; top-30 late sellers R\$224,401.75 = 22.76%; top-10 categories (>=100) R\$637,375.28 = 64.65%. Under a 19% ASSUMED take-rate, implied commission on late GMV = R\$187,325.62. Creates exposure for investigation; cannot establish causality or recoverable amounts.

Slice	Product GMV (R\$)	Share	Comm @19% (ASSUMED)
Total product GMV	13,591,643.70	100%	2,582,412.30
Late product GMV	985,924.34	7.25% of GMV	187,325.62
D1 late GMV	680,097.81	68.98% of late	129,218.58
Top-30 late sellers	224,401.75	22.76% of late	42,636.33
Top-10 cats late GMV	637,375.28	64.65% of late	121,101.30

Financial exposure scenarios | Product GMV = SUM(price) | Late GMV = BRL 985,924 (7.25% of BRL 13,591,644) | Take-rates 15/19/21% UNVERIFIED assumptions — descriptive only

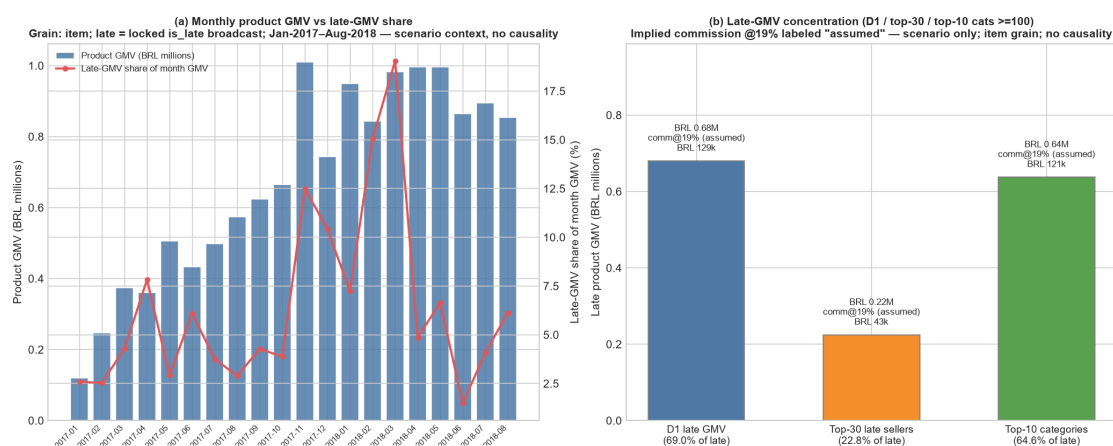


Fig C.6 (Chart 23) Late GMV concentrates in peak months and in the top decile; commission shown is a scenario, not a loss. Technical detail: item grain; comm@19% ASSUMED; no causality.

E. HVIA Solution Proposal

Principle from the brief: "Don't sell a tool. Find a problem worth solving." Every solution below starts from a Section C problem. Build priority follows C.5: core Solutions 1-3 first (strongest evidence, highest reputation stakes), then supporting layers that feed them, then one lighter side solution. E.8 adds Task 2-6 extensions without replacing cores.

E.1 Solution map

#	Problem -> Solution Type HVIA domain
1	P1 Late delivery / reviews -> Predictive Late-Delivery Risk & Response Core Forecasting + Logistics Intelligence
2	P2 Near-zero repeat -> First-Order Retention Risk Flag Core Customer Intelligence
3	P3 Seller Pareto -> Seller Health Score Core Analytics & Decision + AI-Assisted Ops
4	P4 Geo/distance SLA -> Geo-aware Dynamic ETA Supporting (feeds #1) Logistics Intelligence
5	P5 Freight pressure -> Freight-Ratio Catalog Alert Side Analytics & Decision Systems
6	P6 Review-text delivery anger -> Review-Text Triage Supporting (feeds #1,#2) Customer Intelligence + AI Ops
7	P7 Growth/seasonality -> Seasonal Demand Signal Supporting (feeds #1) Forecasting & Predictive + Automation & Data Pipelines (scheduled refresh)

Automation & Data Pipelines layer (feeds all alerts): scheduled ingest of orders/items/reviews, grain-aware late-flag broadcast, and automated risk-list refresh for Solutions 1-3. No new numbers; labeling only.

Lead idea for outreach (one headline): Predictive late-risk early warning with Seller Health as a feature (Solution 1 + 3). The rest below are supporting/side layers. No model was trained or validated in this task.

E.2 Core 1 - Predictive Late-Delivery Risk & Response (P1)

Problem: only 6.8% of delivered orders are late vs ETA, yet average review is lower when late: 4.29 (on-time/early) vs 2.27 (-2.02 stars, descriptive). Under the shared storefront this pattern is consistent with reputation contagion on one Olist marketplace account - not an isolated CX issue.

What it does: at ship-prep time, score each order for probability of missing the *promised* ETA (catch cases where the generous buffer still will not be enough). Limitation: no model was trained or validated in this task.

- Inputs: seller-customer distance (r~0.39 with days); same-state vs cross-state; weight/category as freight-cost and handling proxy (weight-freight r~0.61 describes cost only, not lateness; freight-late co-occurrence is weak r~0.12); seller historical on-time (from Seller Health); current volume load (Seasonal Demand). No carrier ID exists in this extract (orders has only carrier timestamp; items has shipping_limit_date); use freight-per-km proxy or drop; no carrier attribution.
- Actions: Low = standard flow. Medium = proactive customer notice with a revised realistic window before the ETA is missed. High (+ poor Seller Health) = escalate to seller to expedite; flag for priority support if delay still occurs.
- KPI: shrink the on-time vs late review gap (baseline -2.02) for medium-risk notified vs control; secondary: % of high-risk orders still late after escalation (should fall).

Evidence: Charts 04, 05, 06, 15, 19.

E.3 Core 2 - First-Order Retention Risk Flag (P2)

Problem: crude one-time 96.88% buy once; only ~3.1% return in the full window. Growth is acquisition-led; commission LTV would barely compound if repeat stays low (hypothesis; no CAC or customer-value data). Cohort view (eligible H1) shows 4.68% 12m repeat - still low, but a fairer fixed-window KPI than crude alone.

What it does: when a first order completes (delivered + reviewed, or delivered + no review after N days), score repeat-buy probability from that single experience - turning blanket "come back" campaigns into a targeted list worth a proactive touch.

- Inputs: late/high-risk flag from Solution 1; fulfilling seller Health Score; order value/category; review score if given.
- Action: high-value + high-churn-risk customers get a lightweight retention action (support check-in or small proactive gesture) - cost proportional to likely worth.
- KPI: repeat-purchase rate in the flagged high-risk/high-value cohort vs eligible 12m baseline (H1-2017 4.68% on eligible denominator) with a control group from the same cohort; do not use the 3.1% crude full-window rate as the comparator.

Evidence: Charts 08, 22.

E.4 Core 3 - Seller Health Score (P3)

Problem: top 10% of sellers -> ~67.5% product revenue; bottom 50% -> ~3.2%. Under a shared storefront, low GMV does not mean low reputation risk.

What it does: composite per-seller score (late rate, avg review, GMV contribution, growth trend). Tiers e.g. Protect and Grow / Monitor / At-Risk - any size seller whose delivery/review pattern threatens the shared account. Minimum-volume rule: apply the ≥ 10 delivered filter (1,237 sellers) for rate headlines; order-weighted late rates are flat ~5-7% across deciles. Tier thresholds are TBD (not defined here).

- Action: tier feeds Solution 1 as a delay-model feature; account/support attention prioritizes At-Risk sellers instead of spreading evenly across 3,095 merchants.
 - KPI: earlier detection of reputation-risk sellers; falling share of GMV from At-Risk sellers over time as intervention works.
- Evidence: Charts 10, 20. Build note: ship #3 first or alongside #1 (#1 needs the score).

E.5 Supporting components (not standalone pitches)

Scoped as accuracy layers inside Solutions 1-3 - not separate products to sell first.

E.5.1 Geo-aware Dynamic ETA (P4 -> feeds Solution 1)

Replace a flat, buffer-heavy national ETA with a per-route estimate from historical same-state vs cross-state and distance-bucket times (distance<->days $r \sim 0.39$). This is how Solution 1 knows what a realistic promise looks like for a seller-customer pair. Success must be measured by ETA calibration/accuracy and review score, not late rate alone (tightening ETA mechanically raises late rate under late = delivered > ETA). Who sets the ETA (Olist vs marketplace) is not visible in the data. Evidence: Charts 05, 11, 12, 16.

E.5.2 Review-Text Triage (P6 -> feeds Solutions 1 & 2)

Among commented 1-stars ($n=8,744$), delivery/delay keyword hit-rate (unvalidated) $\sim 41\%$ exceeds product/quality $\sim 13\%$, yet median review response is ~ 40 hours. A lightweight classifier routes new delivery-related negatives to a priority support queue and feeds which features Solution 1 should weight. Evidence: Charts 07, 18. Keyword stems are unvalidated; precision unknown.

E.5.3 Seasonal Demand Signal (P7 -> feeds Solution 1)

Volume is highly seasonal (e.g. Nov 2017 peak 7,544 orders). During peaks, baseline late risk is higher independent of one seller or route (descriptive). Feed Solution 1 as a risk multiplier in forecasted high-volume windows. Evidence: Charts 01, 02, 14, 23a.

E.6 Side - Freight-Ratio Catalog Alert (P5)

Freight $\sim 17\%$ of product revenue; item-level freight share of (price+freight) $\sim 21\%$, some categories $\sim 40\%$ (e.g. DVDs/Blu-ray). At catalog add/reprice, flag items above a freight-to-price threshold so merchandising can intervene (bundle, min order, carrier negotiation) before weak conversion. Domain: Analytics & Decision Systems. Evidence: Charts 12, 13, 17, 21. Rank headlines only for cats ≥ 100 delivered orders.

E.7 Suggested build order

Order	What to ship
1	Solutions 1-3 (core). Prefer Seller Health (#3) first or parallel with #1.
2	E.5.1 Geo ETA + E.5.3 Seasonal signal - upgrade Solution 1 accuracy.
3	E.5.2 Review-text triage - strengthens Solutions 1 and 2.
4	E.6 Freight catalog alert - light dependency on the rest.
5	E.8 extensions (delivery-stage, exposure-tier, category-risk, late-GMV + P2 60d) - layer onto cores after baselines.

E.8 Extensions from Tasks 2-6 (same style as E.2-E.6)

These extend existing cores/supports. They do not replace P1-P7 or Solutions 1-3. Wording stays descriptive (shows / indicates / co-occurs / creates exposure / warrants investigation). This report does not establish causality, loss, recoverable amounts, or carrier attribution.

Late-grain note: see definition in C.2 (order 6,534 / item 7,264 / pair 6,547).

E.8.1 Handoff-timer + quarantine (extends E.2 / E.5.1)

What: Monitor seller->carrier handoff clock after approval; quarantine / escalate orders whose handoff age crosses a threshold before transit even starts.

Evidence: see C.2 / Chart 19 (Fig C.2b).

Mechanism: early timer on the stage that shows the widest upper tail, feeding Solution 1 risk and Dynamic ETA realism.

Needs: reliable approved_at + carrier timestamps; DQ handling for negatives; seller contact path.

KPI: share of late orders with handoff > p90 among notified vs control; secondary: review gap on timer-touched orders.

Limitation: co-occurrence only - cannot establish that handoff causes lateness or review collapse; no carrier attribution.

E.8.2 Exposure tier playbook (extends E.4)

What: Playbook that tiers sellers by exposure - D1 / top-30 late-count / thin-volume volatility - instead of GMV-only attention.

Evidence: see C.3-C.4 / Chart 20 (Fig C.3c).

Mechanism: maps Seller Health tiers to ops actions proportional to concentrated late flags and thin-n rate volatility.

Needs: seller-pair late grain labeled; >=10 filter for rate headlines.

KPI: share of pair-late flags under Monitor/At-Risk tiers over time; coverage of late GMV under tiered outreach.

Limitation: concentration indicates exposure, not inefficiency or causal seller blame.

E.8.3 Category-risk lens >=100 only (extends E.6 / E.5.1)

What: Assortment risk view restricted to categories with >=100 delivered orders (51/71 = 97.77% items / 97.63% rev); Unknown excluded from ranks.

Evidence: see C.2 / Chart 21 (Fig C.2c).

Mechanism: feeds Freight Alert + ETA features with n-safe category priors.

Needs: item-grain late broadcast labeled; dual n_items + n_orders on every row.

KPI: % of late items in >=100 ranked cats under review; catalog flags fired.

Limitation: category co-occurs with late flags; cannot establish causality or justify delisting from this evidence alone.

E.8.4 Late-GMV view + peak guardrail (extends E.5.3)

What: Ops dashboard of late product GMV (not order-count alone) with peak-month guardrails when within-month late-GMV share spikes.

Evidence: see C.1 and C.6 / Chart 23 (Fig C.6).

Mechanism: Seasonal Demand multiplier keyed to late-GMV share, not volume only.

Needs: item-broadcast late GMV pipeline; explicit ASSUMED take-rate labels.

KPI: peak-month late-GMV share vs baseline 7.25%; scenario commission on late GMV under stated assumption.

Limitation: scenarios are NOT loss / NOT verified take-rate / NOT recoverable; spikes warrant investigation only.

P2-refined - 60d retention window with 4.68% KPI (extends E.3)

What: Keep crude 96.88% for full-window honesty, and add a fixed-horizon playbook anchored on H1-2017 12m repeat 4.68% with a ~60d median-to-second outreach window (median days to second among 12m repeaters = 60.5d).

Evidence: see C.4 / Chart 22 (Fig C.4c).

Mechanism: Solution 2 timing and success metric use eligible 12m (and ~60d touch) instead of only crude lifetime repeat.

Needs: eligibility cutoff (2017-10-17 style) in production; censoring labeled.

KPI: 12m repeat in treated first-order cohort vs 4.68% H1 baseline (eligible denominator only).

Limitation: cohort rate is not a replacement for crude 96.88%; no claim that fixing delivery raises retention to a target.

One-line pitch (2016-2018 lens): HVIA helps Olist protect the shared marketplace reputation and commission economics by predicting late risk before reviews show lower scores, focusing retention on weak first orders, ranking sellers by health - not by GMV alone - and sizing late-GMV exposure as labeled scenarios.

Bridge to today (hypothesis, not a finding): If mapped to today's Olist, the same ideas would translate to: late-risk scoring for merchant shipments via Pax; merchant health as a possible input to Flip underwriting; ETA calibration inside ERP/Vnda flows. The 2016-2018 evidence does not prove these transfers; they are hypotheses.

References (sources actually opened and checked)

1) Local Olist order/customer/item/review tables (9 CSVs). 2) Analysis script and generated tables/charts (all numbers recomputed locally). 3) Locked definitions appendix (grains/denominators). 4) Public Kaggle dataset page for the Olist extract (dataset identity only). Company facts above are company-reported / approximate / unverified unless labeled; no web company source was opened for this pass.

End of Final Integrated Report (A-B-C-E + Tasks 2-6 evidence; D folded into C). Charts 01-23 with locked definitions appendix. Outreach draft (F) and LinkedIn post remain pending.